

Graphing and Analyzing Hyperbolas

Precalculus / Algebra 2 Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Identify the standard form of a hyperbola and distinguish it from an ellipse
- Find the center, vertices, foci, and asymptotes of a hyperbola
- Graph a hyperbola by constructing the framework, rectangle, and asymptotes

Problems

1. Identify whether the equation below represents an ellipse or a hyperbola. Explain how you know.

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

2. For the hyperbola below, identify the values of a and b.

$$\frac{x^2}{25} - \frac{y^2}{49} = 1$$

3. Find the coordinates of the vertices of the hyperbola below. The center is at the origin.

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

4. Use the formula below to find the value of c for the hyperbola, given a equals 3 and b equals 4.

$$c^2 = a^2 + b^2$$

5. Find the coordinates of the foci for the hyperbola below. The center is at the origin.

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$



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Hyperbolas — Graphing and Analyzing Worksheet

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6. Describe the steps to construct the graphing framework (rectangle) for the hyperbola below, and state the four corner points of the rectangle.

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

7. Write the equations of the two asymptotes for the hyperbola below.

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

8. The hyperbola below has its center at (2, -1). Identify the center, a, and b, then state the vertices.

$$\frac{(x-2)^2}{4} - \frac{(y+1)^2}{9} = 1$$

9. Find the foci of the hyperbola below and write their coordinates.

$$\frac{(x-2)^2}{4} - \frac{(y+1)^2}{9} = 1$$

10. Write the standard form equation of a hyperbola with center at the origin that opens horizontally, with vertices at (6, 0) and (-6, 0), and foci at (10, 0) and (-10, 0).



Graphing and Analyzing Hyperbolas — Answer Key

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Answer Key

1. Answer: Hyperbola, because the terms are subtracted (minus sign between them)

- Check the sign between the two fraction terms.
- A minus sign indicates a hyperbola; a plus sign indicates an ellipse.
- Conclusion: This is a hyperbola.

2. Answer: $a = 5$, $b = 7$

- The denominator under x^2 is a^2 , so $a^2 = 25$, giving $a = 5$.
- The denominator under y^2 is b^2 , so $b^2 = 49$, giving $b = 7$.

3. Answer: Vertices: $(3, 0)$ and $(-3, 0)$

- Since the x^2 term is positive, the hyperbola opens horizontally.
- $a^2 = 9$, so $a = 3$.
- Vertices are located at $(\pm a, 0) = (3, 0)$ and $(-3, 0)$.

4. Answer: $c = 5$

- Substitute $a = 3$ and $b = 4$: $c^2 = 3^2 + 4^2 = 9 + 16 = 25$.
- Take the square root: $c = \sqrt{25} = 5$.

5. Answer: Foci: $(5, 0)$ and $(-5, 0)$

- Find c using $c^2 = a^2 + b^2 = 9 + 16 = 25$, so $c = 5$.
- The hyperbola opens horizontally, so foci are at $(\pm c, 0)$.
- Foci: $(5, 0)$ and $(-5, 0)$.

6. Answer: Corner points: $(3, 4)$, $(3, -4)$, $(-3, 4)$, $(-3, -4)$

- From $x^2 = 9$, we get $x = \pm 3$ (units along x-axis from center).
- From $y^2 = 16$, we get $y = \pm 4$ (units along y-axis from center).
- Draw vertical dashed lines at $x = 3$ and $x = -3$, and horizontal dashed lines at $y = 4$ and $y = -4$.
- The four corners of the rectangle are $(3, 4)$, $(3, -4)$, $(-3, 4)$, and $(-3, -4)$.

7. Answer: $y = (4/3)x$ and $y = -(4/3)x$

- For a horizontal hyperbola centered at the origin, asymptotes have slope $\pm(b/a)$.
- Here $a = 3$ and $b = 4$, so slope = $\pm 4/3$.
- Asymptote equations: $y = (4/3)x$ and $y = -(4/3)x$.

8. Answer: Center: $(2, -1)$, $a = 2$, $b = 3$; Vertices: $(4, -1)$ and $(0, -1)$

- The center is $(h, k) = (2, -1)$.
- $a^2 = 4$, so $a = 2$; $b^2 = 9$, so $b = 3$.



- Since x^2 term is positive, hyperbola opens horizontally.
- Vertices are at $(h \pm a, k) = (2 \pm 2, -1) = (4, -1)$ and $(0, -1)$.

9. Answer: Foci: $(2 + \sqrt{13}), -1$ and $(2 - \sqrt{13}), -1$

- Find c : $c^2 = a^2 + b^2 = 4 + 9 = 13$, so $c = \sqrt{13}$.
- The hyperbola opens horizontally, so foci are at $(h \pm c, k)$.
- Foci: $(2 + \sqrt{13}, -1)$ and $(2 - \sqrt{13}, -1)$.

10. Answer: $x^2 / 36 - y^2 / 64 = 1$

- Since the hyperbola opens horizontally, use the form $x^2/a^2 - y^2/b^2 = 1$.
 - Vertices are at $(\pm 6, 0)$, so $a = 6$ and $a^2 = 36$.
 - Foci are at $(\pm 10, 0)$, so $c = 10$.
 - Use $c^2 = a^2 + b^2$: $100 = 36 + b^2$, so $b^2 = 64$.
 - Standard form: $x^2/36 - y^2/64 = 1$.
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